Thesis - Title

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1 Introduction

1.1 Prediction papers

1.1.1 Segura-Lepe et al. (2019)

Paper Segura-Lepe *et al.* (2019).

Compared predictive models built in **gene space** versus **pathway space**. - Used **PCA** to summarize genes within each pathway; PC1 was used as the pathway score.

- Pathway-based models were generally **more robust to added noise** than gene-level models.
- Surprisingly, **randomized pathways had similar predictive accuracy and robustness** to real biological pathways.
- However, models based on real pathways were **more parsimonious**, with fewer pathways contributing strongly.
- Results were consistent across different datasets and modeling approaches.

1.1.2 Zheng et al. (2020)

Paper by Zheng *et al.* (2020).

- Compared **gene-level** and **pathway-level** models for cancer survival prediction.
- Used TCGA gene-expression data and **MSigDB Hallmark pathways**.
- Used **GSVA** to create sample-specific pathway scores.
- Applied **Lasso-penalized Cox regression** for survival prediction.
- Gene-level and pathway-level models had similar predictive performance on real TCGA data.
- Pathway-level models were generally **more stable, more parsimonious, faster, and easier to interpret**.

- Simulations showed that **inter-gene correlation strongly affects predictive performance**; when correlation was removed, pathway-level models performed much better than gene-level models.

1.1.3 Relevance to This Thesis

These studies suggest that pathway-based modeling can improve robustness, stability, and interpretability, but the results may depend on the **pathway definition, pathway-scoring method, correlation structure, and predictive model**. This motivates comparing different pathway collections, including KEGG, Reactome, GO, WGCNA-derived modules, and randomized pathways, together with scoring methods such as PCA and GSVA.

2 Methods

2.1 Overview

The study will compare different choices at three main stages of pathway-based predictive modeling:

1. **Pathway definition**
2. **Pathway-level scoring**
3. **Predictive modeling**

The complete analysis will be repeated across multiple gene-expression datasets to evaluate whether the results are consistent across different applications.

2.2 Problem formulation

Consider a gene expression dataset collected from n subjects, where expression is measured for p genes. Let

$$\mathbf{X} \in \mathbb{R}^{n \times p}$$

denote the observed gene expression matrix, where X_{ij} is the expression value of gene j for subject i. For each subject $i \in \{1, \ldots, n\}$, let T_i denote the true overall survival time and C_i the censoring time. The observed follow-up time is $U_i = \min(T_i, C_i)$, and the event indicator is $\delta_i = I(T_i \leq C_i)$, where $\delta_i = 1$ indicates an observed death and $\delta_i = 0$ indicates a censored observation. The observed survival outcome is therefore represented by $Y_i = (U_i, \delta_i)$. Let $\mathbf{W}_i$ denote the vector of clinical covariates for patient i, such as age, sex, and disease stage.

Let $\mathcal{G} \subseteq \mathcal{P}(\{1, \ldots, p\})$ denote a collection of pathways, where $\mathcal{P}$ denotes the power set. For a pathway $G \in \mathcal{G}$, the elements of G index the genes belonging to that pathway, and

$$\mathbf{X}(G) \in \mathbb{R}^{n \times |G|}$$

denotes the gene expression matrix restricted to the genes contained in G. Different pathway collections, such as KEGG, Reactome, and Gene Ontology, may define different sets of pathways and different gene memberships within those pathways.

For each pathway $G \in \mathcal{G}$, a pathway-level representation summarizes expression into one or more subject-specific features. Some scoring methods additionally use genes outside the pathway as a reference background. Let

$$\mathbf{Z}_{iG}^{(s)} = \left(Z_{iG1}^{(s)}, \ldots, Z_{iGK_s}^{(s)}\right)^T \in \mathbb{R}^{K_s}$$

denote the pathway-level representation for subject i and pathway G obtained using representation s, where K_s denotes the number of score components in that representation.

2.3 Pathway Collections

Several approaches for defining groups of genes will be considered:

- KEGG : KEGG (Kyoto Encyclopedia of Genes and Genomes) is a curated pathway database that organizes genes and proteins into biological pathways representing cellular processes, metabolism, signaling, and disease mechanisms. KEGG pathways are widely used in genomic studies because they provide interpretable groups of genes with known functional relationships.
- Reactome: Reactome is a curated biological pathway database that describes molecular reactions and signaling events in detail. Its pathways are organized hierarchically, allowing broad biological processes to be divided into more specific subprocesses. Reactome often provides more detailed pathway definitions than KEGG.
- Gene Ontology (GO): Gene Ontology classifies genes according to three major categories: Biological Process, Molecular Function, and Cellular Component. Unlike traditional pathway databases, GO focuses on functional annotations rather than only defined signaling or metabolic pathways. GO gene sets can therefore be large and highly overlapping.
- MSigDB Hallmark Gene Sets: The Hallmark collection contains compact gene sets representing well-defined biological states and processes, such as cell proliferation, inflammation, apoptosis, and immune response. These gene sets were designed to reduce redundancy among larger MSigDB collections and provide more interpretable biological signals.

- Randomized pathway collections: Randomized pathways are artificial gene sets created by shuffling genes while preserving features such as pathway size. They serve as a negative control to test whether real biological pathway structure provides better predictive information than arbitrary gene groupings.
- (Weighted Gene Co-expression Network Analysis) is a data-driven method that groups genes into modules based on similar expression patterns across samples. Unlike curated pathway databases such as KEGG or Reactome, WGCNA does not rely on predefined biological knowledge; instead, it identifies groups of genes that tend to be co-expressed in the observed dataset. Each module can be summarized by a module eigengene, which represents the main expression pattern of that group. In this study, WGCNA-derived modules can be treated as an alternative pathway collection and compared with curated and randomized gene sets.

2.4 Pathway-Level Scoring

For each pathway collection, gene-expression measurements will be summarized into patient-specific pathway scores. We will compare single-score representations, including PC1, GSVA, and ssGSEA, with vector representations containing multiple principal components or combinations of scoring methods. These representations will be used as predictors in the survival models described below.

2.4.1 Principal Component Analysis (PCA)

Principal component analysis (PCA) summarizes correlated variables using linear combinations that explain the greatest variation in the data (Jolliffe and Cadima, 2016). In this study, PCA is applied separately to the genes within each pathway after centering and scaling each gene using its training mean and standard deviation.

PCA identifies eigenvectors of the training covariance matrix, which provide the gene weights, called loadings. The corresponding eigenvalues quantify the variance explained by each component. For patient i, the score for principal component k of pathway G is calculated as TODO: notations

$$Z_{iGk}^{(\mathrm{PCA})} = \sum_{g \in G^*} \widetilde{X}_{ig} \hat{v}_{g,Gk},$$

where G^* denotes the set of retained genes in pathway G, $\widetilde{X}_{ig}$ is the standardized expression of gene g in patient i, and $\hat{v}_{g,Gk}$ is the loading of gene g on component k for pathway G.

Standardized expression is defined as

$$\widetilde{X}_{ig} = \frac{X_{ig} - \hat{\mu}_g}{\hat{\sigma}_g},$$

where X_{ig} is the observed gene-expression value, and $\hat{\mu}_g$ and $\hat{\sigma}_g$ are the mean and standard deviation of gene g estimated from training patients. The loadings are also estimated from the standardized training expression data. The hat denotes an estimated quantity. Held-out patients are scored using the same training means, standard deviations, and loadings.

(TODO: Connect this to the previous notations)The eigenvector corresponding to the largest eigenvalue provides the loadings for PC1. The ratio of this eigenvalue to the sum of all eigenvalues gives the proportion of variance explained by PC1. Using these loadings, PC1 provides one pathway score per patient. Alternatively, scores from multiple components, such as PC1, PC2, and PC3, can be combined into a pathway-score vector.

TODO:(add extreme cases)

PCA is implemented using `prcomp()` with `center = TRUE` and `scale. = TRUE`. Held-out patients are scored using the training means, standard deviations, and loadings. Survival outcomes are not used to construct the scores.

2.4.2 Gene Set Variation Analysis (GSVA)

Gene set variation analysis (GSVA) is an unsupervised method that summarizes gene expression into one enrichment score per patient and pathway. Unlike PCA, GSVA uses a rank-based approach to describe the relative enrichment of pathway genes.

First, each gene's expression is transformed using an estimated cumulative distribution across patients. For continuous expression data, this distribution is estimated using a Gaussian kernel. The transformed values are then ranked within each patient and converted into rank-based weights. For continuous gene-expression data, GSVA estimates each gene's cumulative distribution function using a Gaussian kernel. For gene g in patient i, the transformed expression is

$$U_{ig} = \widehat{F}_g(X_{ig}) = \frac{1}{n}\sum_{j=1}^{n} \int_{-\infty}^{(X_{ig}-X_{jg})/h_g} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{t^2}{2}\right) dt,$$

where X_{ig} is the expression of gene g in patient i, n is the number of patients in the batch being scored, and h_g is the gene-specific bandwidth controlling the smoothness of the distribution estimate. For the Gaussian kernel used by GSVA, $h_g = s_g/4$, where s_g is the standard deviation of gene g across these patients.

Equivalently, TODO: move or erase

$$U_{ig} = \frac{1}{n} \sum_{j=1}^{n} \Phi \left(\frac{X_{ig} - X_{jg}}{h_g} \right),$$

where $\Phi(\cdot)$ is the standard normal cumulative distribution function. The value U_{ig} describes the relative position of patient i's expression within the estimated distribution for gene g.

For patient i and pathway G, let $g_{i1}, \ldots, g_{ip}$ denote the ordered genes, where p is the total number of eligible genes. At position ℓ in this ordering, define the running difference as

$$D_{iG}(\ell) = P_{iG}^{\text{hit}}(\ell) - P_{iG}^{\text{miss}}(\ell),$$

where

$$P_{iG}^{\text{hit}}(\ell) = \frac{\sum_{r=1}^{\ell} w_{ir} \, \mathbb{I}(g_{ir} \in G^*)}{\sum_{r=1}^{p} w_{ir} \, \mathbb{I}(g_{ir} \in G^*)},$$

and

$$P_{iG}^{\text{miss}}(\ell) = \frac{\sum_{r=1}^{\ell} \mathbb{I}(g_{ir} \notin G^*)}{p - |G^*|}.$$

Here, G^* denotes the retained genes in pathway G, w_{ir} TODO: is the nonnegative rank-based weight assigned to gene g_{ir}, and $\mathbb{I}(\cdot)$ is the indicator function, equal to one when its condition holds and zero otherwise. Thus, $P_{iG}^{\text{hit}}(\ell)$ is the cumulative normalized weight of pathway genes encountered up to position ℓ, while $P_{iG}^{\text{miss}}(\ell)$ is the proportion of non-pathway genes encountered up to that position.

TODO: move Using the difference-based scoring option (`maxDiff = TRUE`, `absRanking = FALSE`), the GSVA score is

$$Z_{iG}^{(\text{GSVA})} = \max_{0 \le \ell \le p} D_{iG}(\ell) + \min_{0 \le \ell \le p} D_{iG}(\ell),$$

with $D_{iG}(0) = 0$. This is the largest positive excursion minus the magnitude of the largest negative excursion. A positive score indicates a larger positive excursion, whereas a negative score indicates a larger negative excursion.

GSVA is implemented using the R package `GSVA`. Because the expression-distribution transformation depends on the patients included in the calculation, GSVA scores are cohort-dependent. Training and held-out batches are scored separately, and any subsequent score standardization uses training means and standard deviations.

(TODO: add extreme case both PCA and GSVA) The resulting scalar or vector pathway representations will be used in direct and two-stage modeling approaches. Pathway-score centering and scaling will use estimates from the appropriate training folds. Predictive performance will be assessed using held-out predictions, as described in Section 2.5 and Section 2.6.

2.4.3 Scalar and Vector Pathway Representations

Using the notation introduced above, scalar representations have $K_s = 1$, whereas vector representations have $K_s > 1$.

Scalar representations will include the first principal component (PC1), GSVA, and single-sample gene set enrichment analysis (ssGSEA), considered separately. Each produces one score per patient for each pathway.

Vector representations will include multiple principal components, such as $(PC1, PC2, PC3)$, and combinations of scoring methods, such as $(PC1, GSVA, ssGSEA)$.

Additional candidate representations will summarize expression across genes within each pathway for each patient. These may include the mean ($\bar{x}_{iG}$), standard deviation (s_{iG}), and the difference between the mean and median, giving

$$\mathbf{Z}_{iG}^{(\text{summary})} = \left(\bar{x}_{iG}, s_{iG}, \bar{x}_{iG} - \text{median}_{g \in G}(X_{ig})\right)^T.$$

Here, the mean, standard deviation, and median are calculated across genes in pathway G within patient i, and X_{ig} denotes the expression of gene g in patient i. The expression scale and any gene standardization will be specified before calculating these summaries.

These representations will be compared to assess whether retaining multiple features per pathway improves survival prediction relative to using a single pathway score.

2.5 Predictive Models

2.5.1 Cox Proportional Hazards Model

The Cox proportional hazards model will be used as a baseline survival model. For patient i, the hazard function is

$$h(t \mid \mathbf{x}_i) = h_0(t) \exp(\mathbf{x}_i^T \beta),$$

where $h_0(t)$ is the baseline hazard function, $\mathbf{x}_i$ represents the pathway-level predictors, and β contains the corresponding regression coefficients.

The Cox model relates the pathway-level predictors to the instantaneous risk of experiencing the event. The exponentiated regression coefficient,

$$\exp(\beta_j),$$

is interpreted as a hazard ratio. A hazard ratio greater than 1 indicates an increased hazard, whereas a hazard ratio less than 1 indicates a decreased hazard, holding the other predictors constant.

An important assumption of the Cox model is the proportional hazards assumption, which states that the hazard ratio between individuals remains constant over time.

2.5.2 Accelerated Failure Time Model

The accelerated failure time (AFT) model will also be considered as an alternative parametric approach for modeling overall survival. Unlike the Cox proportional hazards model, which models the hazard of an event, the AFT model directly models survival time.

The general AFT model can be written as

$$\log(T_i) = \beta_0 + \mathbf{x}_i^T \beta + \sigma \epsilon_i,$$

where T_i represents the true, potentially unobserved survival time of patient i, $\mathbf{x}_i$ represents the pathway-level predictors, β contains the regression coefficients, σ is a scale parameter, and ϵ_i represents the model error term whose distribution depends on the chosen AFT model.

Exponentiating a regression coefficient gives the time ratio,

$$\exp(\beta_j).$$

A time ratio greater than 1 indicates that survival time is multiplied by a factor greater than 1 as the predictor increases, whereas a time ratio less than 1 indicates shorter survival time.

2.6 Performance Evaluation Metrics

2.6.1 Censoring-adjusted Root Mean Squared Error (RMSE)

For survival models using GSVA pathway scores, prediction error in survival time can be evaluated using the inverse probability of censoring weighted RMSE described by Grimes et al. (2018), Equation (4). GSVA supplies the pathway predictors; the fitted survival model supplies a predicted survival time for each held-out patient.

Using the observed follow-up time U_i and event indicator δ_i defined above, let $\widehat{T}_i$ denote the predicted survival time. Let $\widehat{G}(t)$ estimate the probability of remaining uncensored beyond time t, obtained by fitting a Kaplan–Meier estimator with censoring treated as the event. For n_{test} held-out patients, the adjusted RMSE is

$$\widehat{\text{RMSE}}_{\text{IPCW}} = \left[\frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} \frac{\delta_i \left(U_i - \widehat{T}_i \right)^2}{\widehat{G}(U_i^-)} \right]^{1/2}.$$

Here, $\widehat{G}(U_i^-)$ is the censoring survival probability immediately before the observed event time. Only observed deaths contribute squared errors; their inverse probability weights account for incomplete follow-up. The denominator remains the total number of test patients, rather than the number of observed deaths or the sum of weights. This estimator assumes independent censoring under the chosen censoring model and positive censoring survival probabilities at the event times evaluated.

Lower values indicate more accurate survival-time predictions. RMSE has the same units as survival time: for TCGA-LUAD follow-up recorded in days, the result is expressed in days. Unlike the Brier score, this metric evaluates a predicted time rather than a survival probability. A scalar time prediction must therefore be specified consistently, such as the conditional median survival time. For a Weibull AFT model, this median can be obtained from the fitted survival distribution. A Cox linear predictor alone is a risk score, so a Cox time prediction additionally requires the estimated baseline survival curve; its median may be unavailable if that curve does not reach a survival probability of 0.5.

RMSE evaluates the survival model using GSVA predictors, rather than the GSVA scoring procedure itself. The same metric can also evaluate models using PCA predictors when comparable survival-time predictions are available. Adding this definition to the thesis does not mean RMSE has already been calculated in the current Reactome results.

2.6.2 Concordance Index (C-index)

Uno *et al.* (2011) Model discrimination will be evaluated using Uno's concordance index (C-index). The C-index measures the ability of a survival model to correctly rank patients according to their predicted risk. A pair is concordant when the patient with the shorter survival time is assigned a higher predicted risk.

Because survival outcomes may be right-censored, Uno's C-index will be used. Unlike the conventional C-index, Uno's estimator uses inverse probability of censoring weighting (IPCW) to account for censoring.

For a prespecified time horizon τ, Uno's C-index is estimated as

$$\widehat{C}_\tau = \frac{\sum_{i=1}^{n}\sum_{j=1}^{n}\frac{\delta_i}{\widehat{G}(U_i)^2} I(U_i < U_j,\ U_i < \tau) I(\hat{\beta}^T Z_i > \hat{\beta}^T Z_j)}{\sum_{i=1}^{n}\sum_{j=1}^{n}\frac{\delta_i}{\widehat{G}(U_i)^2} I(U_i < U_j,\ U_i < \tau)}.$$

Here, U_i is the observed survival time for patient i, δ_i is the event indicator, $\widehat{G}(U_i)$ is the estimated probability of remaining uncensored at time U_i, and $\hat{\beta}^T Z_i$ is the predicted risk score. The indicator function $I(\cdot)$ equals 1 when the specified condition is satisfied and 0 otherwise. Higher C-index values indicate better discrimination.

The time horizon τ will be selected to avoid unstable estimation in the tail of the survival distribution, where relatively few patients may remain under observation.

2.6.3 Brier Score

The Brier score evaluates the accuracy of predicted survival probabilities at a specific time point. For right-censored survival data, inverse probability of censoring weighting (IPCW) is used to account for censoring (Graf *et al.*, 1999; Gerds and Schumacher, 2006).

At time t, the censoring-adjusted Brier score is

$$\widehat{BS}(t) = \frac{1}{n}\sum_{i=1}^{n}\left[\frac{I(U_i \le t, \delta_i = 1)}{\widehat{G}(U_i)}\left(0 - \widehat{S}_i(t)\right)^2 + \frac{I(U_i > t)}{\widehat{G}(t)}\left(1 - \widehat{S}_i(t)\right)^2\right].$$

where:

- n is the total number of patients.
- U_i is the observed survival or follow-up time for patient i.
- δ_i is the event indicator, where $\delta_i = 1$ indicates that the event was observed and $\delta_i = 0$ indicates censoring.
- t is the time point at which prediction error is evaluated.

- $\widehat{S}_i(t)$ is the predicted probability that patient i survives beyond time t.
- $\widehat{G}(t)$ is the estimated probability of remaining uncensored beyond time t.
- $I(\cdot)$ is an indicator function that equals 1 when the specified condition is satisfied and 0 otherwise.

The first term represents patients who experience the event by time t, while the second term represents patients who are known to survive beyond time t. The inverse probability of censoring weights adjust for incomplete observation due to censoring. A lower Brier score indicates lower prediction error.

2.6.4 Integrated Brier Score (IBS)

The integrated Brier score (IBS) summarizes prediction error over a specified time interval rather than at a single time point (Graf *et al.*, 1999; Gerds and Schumacher, 2006).

$$IBS(\tau) = \frac{1}{\tau} \int_0^\tau \widehat{BS}(t)\, dt.$$

where:

- $\widehat{BS}(t)$ is the Brier score at time t.
- τ is the upper limit of the evaluation time interval.
- $\int_0^\tau \widehat{BS}(t)\, dt$ represents the accumulated prediction error over the interval from 0 to τ.

Thus, the IBS represents the average prediction error over the specified time interval. Lower IBS values indicate better overall predictive accuracy.

2.7 Model Fitting and Evaluation

Predictive performance will be evaluated using k-fold cross-validation. The data will be divided into k approximately equal-sized folds. In each iteration, $k-1$ folds will be used as the training set and the remaining fold will be used as the test set. This process will be repeated until each fold has served once as the test set.

The same cross-validation partitions will be used across all modeling approaches to ensure a fair comparison. Model fitting and any data-dependent estimation will be performed using only the training data within each fold to avoid information leakage. The fitted models will then be used to generate predictions for patients in the corresponding held-out test fold.

Models will be fitted for each combination of pathway collection, pathway-level scoring method, and predictive model described in the previous sections. For each pathway collection, pathway-level scores will first be generated using the scoring methods described in Section 2.4. These scores will then be used as predictors in the models described in Section 2.5.

For example, combinations may include **KEGG + PCA + Cox, PH KEGG + GSVA + AFT**, **Reactome + PCA + Cox PH** and **Reactome + GSVA + AFT**. The same procedure will be repeated systematically across the specified pathway collections, pathway-scoring methods, predictive models, and datasets.

Predictive performance will be assessed using the performance metrics described in Section 2.6. Performance will be calculated using predictions from the held-out observations so that model evaluation is based on data that were not used for model fitting.

2.8 Results

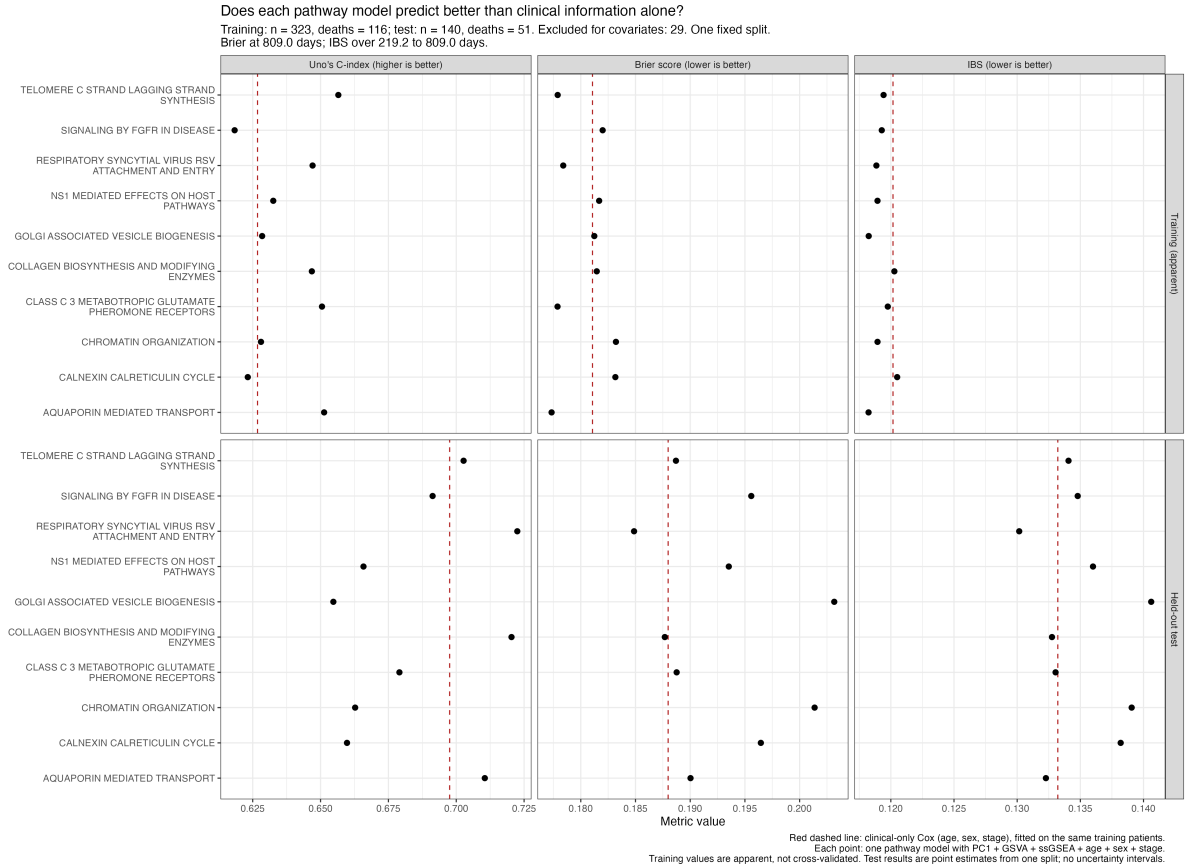


Figure 1: Training and held-out test performance of pathway-specific Cox models compared with clinical-only Cox. Red dashed lines indicate clinical-only performance. Training metrics are apparent; test estimates come from one split.

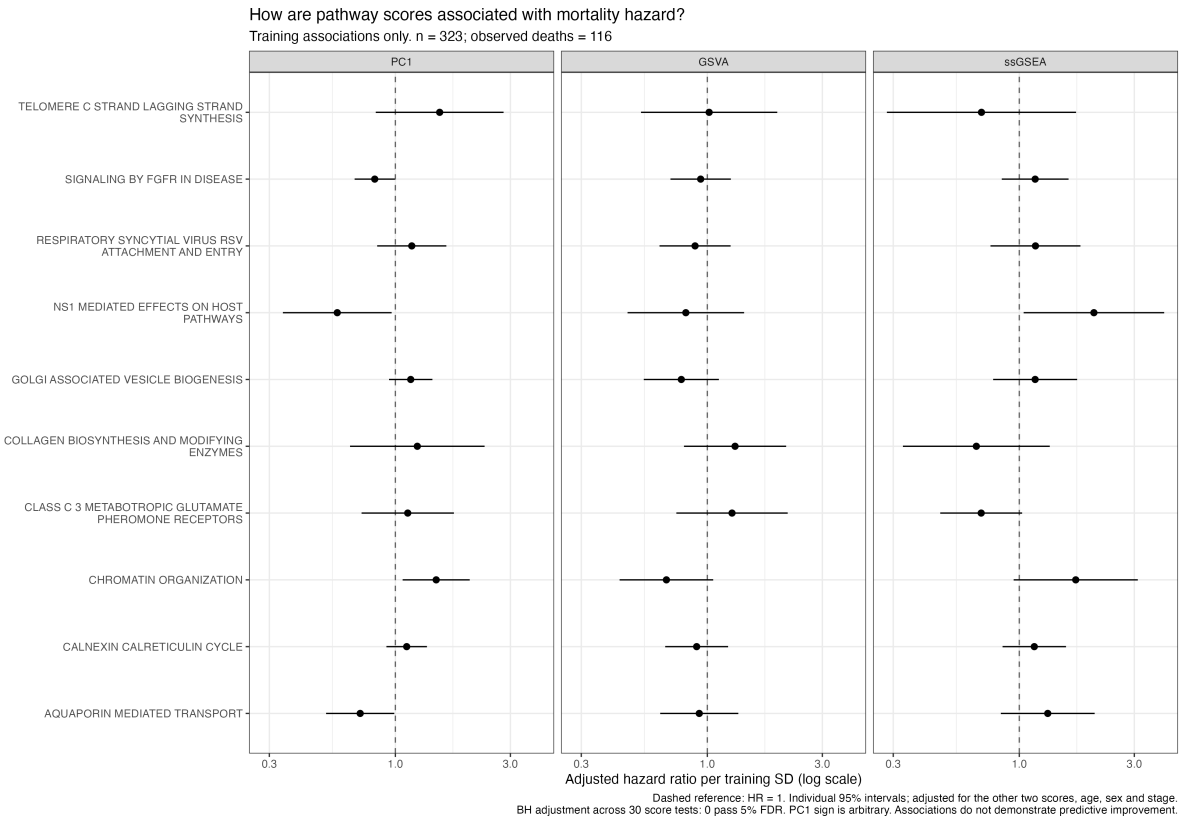


Figure 2: Adjusted associations between pathway scores and mortality hazard in training patients. Points show hazard ratios and lines show individual 95% confidence intervals.



Figure 3: Correlations between PC1, GSVA, and ssGSEA scores across training patients.

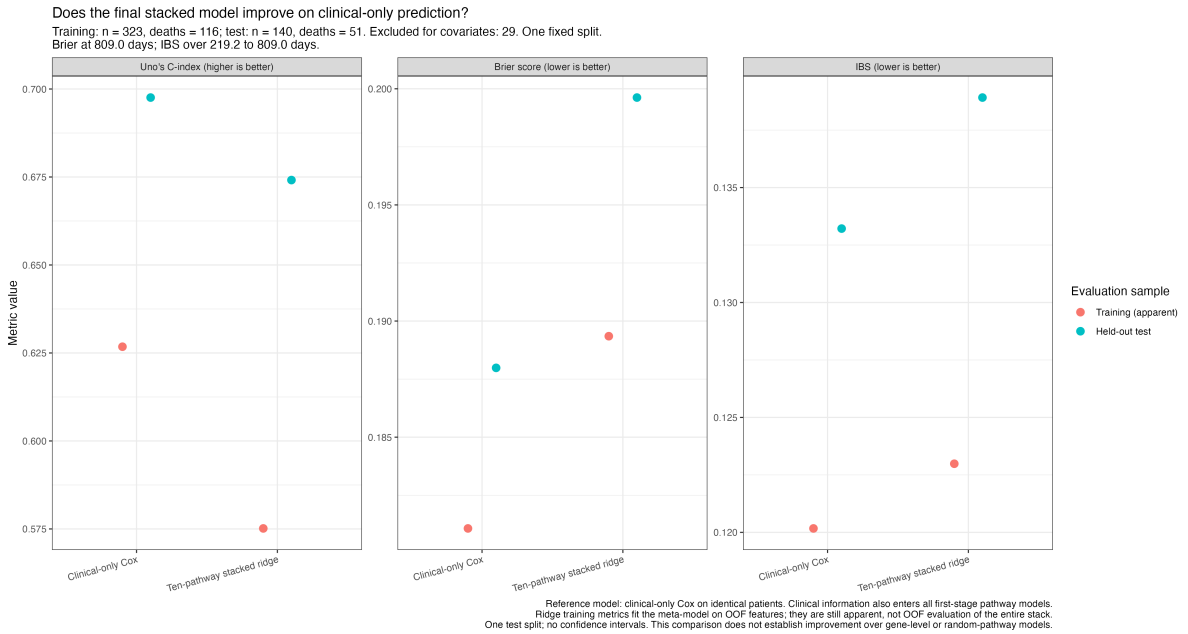


Figure 4: Training and held-out test performance of the stacked ridge model compared with clinical-only Cox. Training metrics are apparent, even though ridge uses out-of-fold pathway features.

2.8.1 Penalized Regression

Three penalized regression approaches will be compared:

- **LASSO regression**, which can shrink some coefficients exactly to zero and therefore perform variable selection.
- **Ridge regression**, which shrinks coefficients toward zero but generally retains all predictors.
- **Elastic Net**, which combines the LASSO and ridge penalties.

These methods will allow comparison of different approaches for handling a potentially large number of correlated pathway predictors.

2.8.2 Random Forest

Random forest models will be used as a tree-based alternative to regression models. Random forests can capture nonlinear relationships and interactions between pathway scores without requiring a prespecified linear relationship.

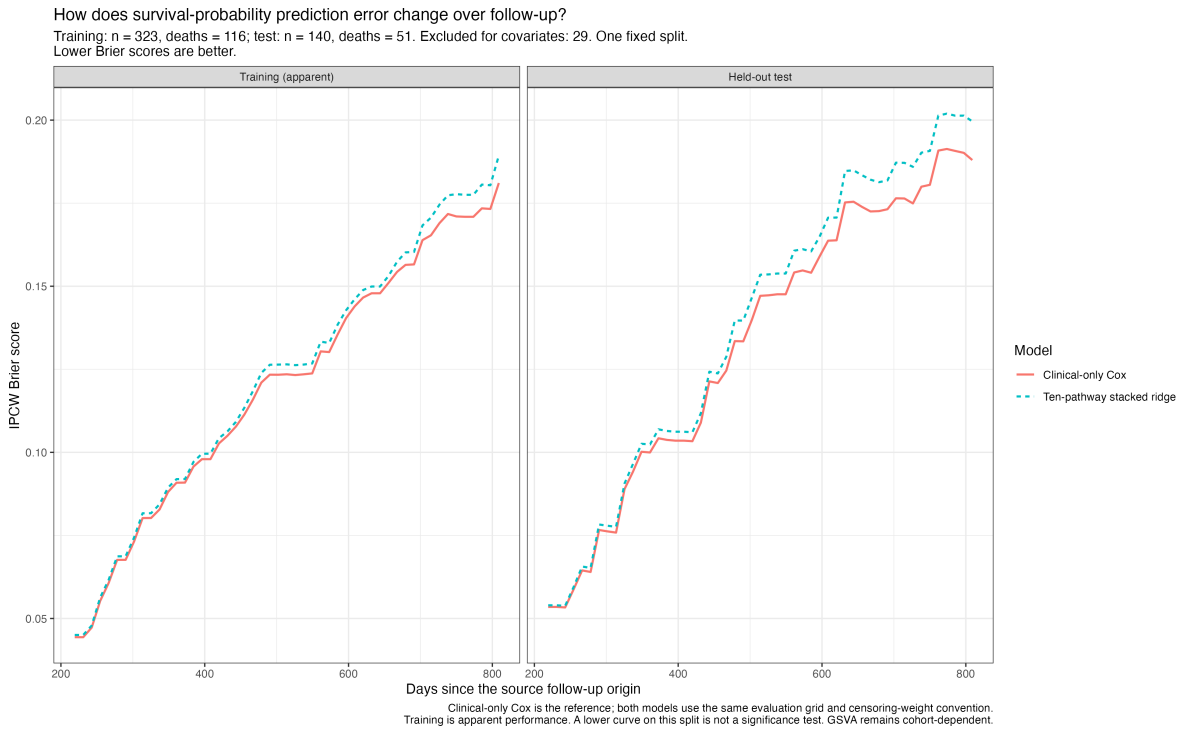


Figure 5: Brier-score curves for stacked ridge and clinical-only Cox models, shown separately for training and test patients. Lower values indicate lower prediction error.

2.8.3 XGBoost

XGBoost will also be evaluated as a boosting-based machine-learning method. XGBoost sequentially combines decision trees to improve predictive performance and can model nonlinear relationships and interactions among pathway-level predictors.

2.9 Model Evaluation

A consistent cross-validation framework will be used to compare all methods fairly. The same training and testing samples will be used when comparing pathway collections, scoring methods, and predictive models.

Where necessary, tuning parameters will be selected using cross-validation within the training data.

The models will be compared using measures of:

- predictive performance,
- model stability,
- model complexity or number of selected predictors,
- computational efficiency,
- and interpretability.

2.10 Comparison Across Methods

The final comparison will consider combinations of:

- pathway collection,
- pathway-scoring method,
- predictive model,
- and dataset.

References :

Gerds,T.A. and Schumacher,M. (2006) Consistent estimation of the expected brier score in general survival models with right-censored event times. *Biometrical Journal*, **48**, 1029–1040.

Graf,E. *et al.* (1999) Assessment and comparison of prognostic classification schemes for survival data. *Statistics in Medicine*, **18**, 2529–2545.

Jolliffe,I.T. and Cadima,J. (2016) Principal component analysis: A review and recent developments. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, **374**, 20150202.

Segura-Lepe,M.P. *et al.* (2019) Predictive modelling using pathway scores: Robustness and significance of pathway collections. *BMC bioinformatics*, **20**, 543.

Uno,H. *et al.* (2011) On the c-statistics for evaluating overall adequacy of risk prediction procedures with censored survival data. *Statistics in Medicine*, **30**, 1105–1117.

Zheng,X. *et al.* (2020) Comparison of pathway and gene-level models for cancer prognosis prediction. *BMC bioinformatics*, **21**, 76.